

Spectra Systems Plan

A focused business plan to operationalize two products: 1) **Civic Indicators Dashboard** for risk intelligence and public accountability. 2) **Offline QA Tutor** for resilient knowledge access and local education.

This plan is structured for direct execution. It includes a crisp narrative, market and customer definitions, pricing, delivery, tech architecture, security, KPIs, and a 30–60–90 day rollout.

1) Executive Summary

Vision: Give communities verifiable signal and resilient knowledge so they can adapt, organize, and thrive under pressure.

Products: - **Indicators Dashboard:** Self-updating, auditable, shareable. Tracks hate incidents, political violence risk, policy shocks, AI weaponization, and social stability metrics. - **Offline QA Tutor:** Local-first RAG assistant that answers with citations from a curated corpus. Delivers essential skills and curriculum without internet.

Why now: Polarization, information collapse, and API instability have created demand for trustworthy, local-first tools with provenance.

Business model: Dual track. Paid subscriptions and service contracts for institutions, plus community deployments with grants and sponsorships.

First target segments: City offices, school districts, faith and cultural centers, local newsrooms, legal aid clinics, healthcare systems, public libraries, and university labs.

2) Products and Offers

A. Civic Indicators Dashboard

Core capabilities - Automated ingestion across official data, vetted media, and NGO trackers. - Normalization and de-duplication. Transparent source log for every datapoint. - Signal detection: rolling z-scores, change point detection, anomaly flags. - Alerts: email, SMS, Signal, RSS, and webhook. - Public UI: maps, timelines, indicator cards, CSV export, and simple API.

Starter indicator set 1. Hate incident rate per 100k with 90-day baseline. 2. Violent crime 7-day vs 30-day average. 3. Law enforcement procurement and deployment notices. 4. Policy shocks that affect civil liberties. 5. AI weaponization signals from awarded contracts and vendor releases. 6. Youth labor and fertility deviations from trend. 7. Narrative amplification index across monitored channels.

Packaging - Free tier: Last 30 days, daily refresh, 3 indicators, public widgets with link attribution. - **Pro:** Hourly refresh, full indicator set, custom alerts, CSV and API, 3 admin seats. - **Enterprise:** Custom sources, on-prem or VPC, SSO, SLAs, audit reports, training.

B. Offline QA Tutor

Core capabilities - Curated corpora: OpenStax, first aid, WHO field guides, Wikipedia, local law basics, vocational manuals. - Local embeddings and quantized models on commodity hardware. - RAG with strict citation and confidence thresholds. - Training mode, lesson packs, and saved threads for classrooms or clinics.

Packaging - Community Kit: Single device, up to 25 concurrent sessions, quarterly corpus updates. - **School Kit:** Lab deployment, device management, lesson packs, annual training. - **Clinic Kit:** First aid and public health focus, bilingual packs, compliance guidance.

3) Customer Profiles and Jobs To Be Done

- **City offices:** Monitor public safety signals, brief councils, respond faster.
- **Newsrooms:** Verify trends and embed live widgets to increase trust.
- **Faith and cultural centers:** Coordinate safety and mutual aid with data.
- **Libraries and schools:** Provide offline learning, skill training, and tutoring.
- **Legal aid and advocacy:** Document patterns, escalate cases, and inform policy.
- **Healthcare systems:** Track social risk, offer offline patient education.

Buying triggers: visible incidents, grant cycles, new admin priorities, curriculum modernization, or a recent failure of cloud tools.

4) Competitive Landscape

- **Dashboards:** Mix of closed vendor suites and single-issue trackers. Gaps: provenance, locality, and community focus.
- **Tutors:** Cloud LLM tools lack offline reliability and citations. Gaps: resilience and verifiable sources.

Spectra edge: local-first design, provenance by default, and clean UX that non-technical users can adopt in under one hour.

5) Pricing and Revenue

Civic Indicators Dashboard - Free: \$0 - Pro: \$149 per month per city or district - Enterprise: \$1,500 per month per region, plus setup fee \$5,000 to \$15,000

Offline QA Tutor - Community Kit: \$1,200 hardware + \$99 per month for updates and support - School Kit: \$7,500 per site per year, includes training and curriculum packs - Clinic Kit: \$4,000 per site per year, includes bilingual health packs

Services - Deploy and train: \$2,500 per engagement - Custom corpus curation: \$2,000 to \$10,000 depending on scope - Grants and sponsorship placement: 10 percent of award to cover overhead

6) Go To Market

Channels - Direct to city managers, chiefs of staff, and council offices. - Partnerships with libraries, school districts, and faith networks. - Press kits for local journalists and nonprofit coalitions. - Webinars and in-person demos.

Assets - One-page briefs with screenshots and example alerts. - Live demo environment with synthetic data and a real big-city pilot. - Case studies after first 60 days of operation.

Motion - Start with a Los Angeles metro pilot and an Austin metro pilot for contrast. - Publish weekly "Spectra Brief" that highlights three indicator moves, with citations. - Offer an install of the Offline Tutor to each pilot partner to create dual value.

7) Delivery and Ops

Team roles - Product lead, ingestion engineer, data analyst, civic partner manager, operations.

Operational rhythms - Weekly release train and indicator review. - Monthly corpus refresh and tutor QA. - Quarterly partner council to adjust indicators and lesson packs.

Support - Email, Signal channel for priority partners, and a public status page.

8) Tech Architecture

Dashboard - Ingest: Python workers with RSS, APIs, and scrapers. - Storage: Postgres with Timescale for time series. - Index: Meilisearch or Elasticsearch for text. - Processing: scheduled jobs and anomaly detection. - API: FastAPI. - Frontend: React, Tailwind, and VegaLite. - Packaging: Docker Compose for single host, Kubernetes optional.

Offline QA Tutor - LLM runtime: llama.cpp with Llama or Mistral small models. - Vector store: Chroma or FAISS. - Embeddings: all-MiniLM L6 v2 running locally. - Orchestration: FastAPI service with retrieval, ranking, and generator. - UI: lightweight React client, offline cache, and export.

Security - No PII collection by default. - Source hashing, signed manifests for corpus bundles. - Role based access for Pro and Enterprise.

9) Data Governance and Ethics

- Publish a data provenance policy with source lists.
 - Provide clear right to contest and remove.
 - Document known limitations and bias risks.
 - Add a public change log and versioned schema.
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10) KPIs

Acquisition - Pilot signups per month - Demo to contract conversion rate

Engagement - Active users per week - Alerts acknowledged within 24 hours - Tutor sessions per device per day

Trust - Percentage of alerts with three or more sources - Average response time for data correction

Impact - Documented policy actions triggered - Classroom hours delivered offline

Financial - Monthly recurring revenue - Gross margin

11) Roadmap

Quarter 1 - MVP dashboard with 5 indicators and alerting - Offline Tutor Community Kit with three corpora - LA and Austin pilots

Quarter 2 - API and widgets - School and Clinic Kits - Grants and press partnerships

Quarter 3 - Enterprise deployments and on-prem option - Custom indicator designer - International corpora add-on

12) Risks and Mitigations

- **Data reliability:** Cross-source validation and human analyst review for critical alerts.
 - **Legal exposure:** Clear terms, public provenance, and counsel review for scraping and public records.
 - **Adoption friction:** One-hour install playbooks and video walkthroughs.
 - **Model drift:** Monthly evaluation sets and regression tests.
 - **Funding gap:** Early revenue from services, small grants, and strategic sponsorships.
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13) 30–60–90 Day Action Plan

Days 1–30 - Lock indicator schema and alert templates. - Build MVP ingest for two cities and three NGO sources. - Assemble Tutor corpus v0 with first aid, OpenStax Algebra, and Wikipedia essentials. - Draft one-page brief and demo video.

Days 31–60 - Run two pilots. Publish weekly Spectra Brief. - Add widgets and CSV export. Add Tutor lesson packs v1. - Close first three Pro subscriptions or one Enterprise.

Days 61–90 - Harden pipeline. Add policy shock detector and procurement watcher. - Ship School and Clinic Kits. Publish first case study and ROI estimates. - Prepare on-prem option for one Enterprise prospect.

14) Assets to Generate Now

- Indicator JSON schema and alert card format.
 - Docker Compose files for both products.
 - Demo scripts and screenshots.
 - Outreach email templates for city offices and schools.
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15) Immediate Next Step

Pick one of these: 1. Convert this plan into a Notion workspace with tasks, owners, and deadlines. 2. Generate the indicator schema and Docker Compose templates. 3. Draft the outreach emails and a two-page pitch.

Reply with your choice and I will produce it now.